

sun visor to become the heads-up display, with a laser projecting driving instructions and maps that appear to float above the road about three meters away," said a *PC World* report.

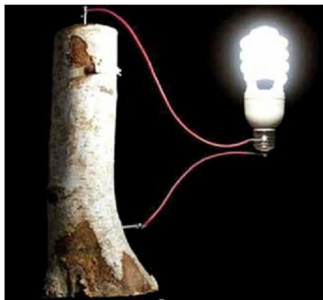
A camera, which is mounted near the rear view mirror, feeds video to a traditional LCD screen mounted on the dashboard after scanning the road ahead. "The system overlays the video with graphics, using object recognition to detect and highlight the nearest vehicle ahead as well as upcoming street signs and traffic stops," the report added. The dashboard screen is also capable of pointing out landmarks including restaurants and gas stations, and adding large icons that float in the air.

A battery made out of wood

In an era where it is cool to be environmentally aware, scientists at the University of Maryland have come up with a formula for batteries that can be counted as cool. These batteries have wood as a major element in their formation, besides fiber and sodium, making them sodium-powered batteries.

"The inspiration behind the idea comes from the trees. Wood fibers that make up a tree once held mineral-rich water, and so are ideal for storing liquid electrolytes, making them not only the base but an active part of the battery," says Hu, an assistant professor of materials science, at the university.

Sodium-ion batteries are, incidentally, considered less powerful and efficient than their lithium-ion counterparts. That doesn't however mean that these batteries can't be useful. According to the team of these scientists, these wooden batteries might not end up within your smartphone or your laptop or in any other fancy gadget, but they are extremely useful for storing power on a large scale basis, given the fact that they are good for storing electrolytes as they can swell and contract many times over.



Battery model made out of wood

Body heat powers flashlight

A 15-year-old school student, Ann Makosinski, has developed an interesting project 'Flashlight Powered By Body Heat,' which has been shortlisted for entry into the final rounds of the Google Science Fair 2013 that received innovative projects from young, high-school students from over 200 countries.



Ann Makosinski with her flashlight

Thermoelectric effect is the soul of the flashlight. It refers to the direct conversion of temperature difference into electricity. Voltage difference is created between the two sides of it at different temperatures with the help of a thermoelectric device. A hollow pipe was covered with aluminium tiles. When air flows through the hollow pipe, it maintains the temperature difference by keeping the other side cool.

As soon as you hold the flashlight, the body heat causes the metallic strip to have higher temperature than the one with aluminium strips. The difference in the temperature is turned into electricity in order to power the LED flashlight with the help of the thermoelectric device.

Flexible 'electronic skin' that can detect touch

The day is not far when amputees would be able to feel changes in their environment through prosthetic limbs. Researchers at the Technion-Israel Institute of Technology have reportedly made a breakthrough using tiny gold particles and a special resin to develop a new type of flexible sensor. The sensor could be integrated into the artificial skin allowing it to detect touch, humidity and temperature as well. The research team said that the flexible sensor can be built at a low cost.

According to lead researcher Hossam Haick, the e-skin was "at least 10 times more sensitive in touch" than any other existing touch-based e-skin systems. It is pertinent to note here that it's the first artificial skin sensor to record temperature and humidity.

'Click' and capture smell instead of images

Believe it or not, but British researchers have developed a camera which can capture smell. This new camera is capable of recording smells instead of images. According to *Guardian*, designer Amy Radcliffe at Central Saint Martins has invented this gadget which is named Madeleine. The gadget could be used for providing meaningful sensory dimension to storing our emotional memories.

"The sense of smell has a direct link to emotional memory. It is the sense we react to most instinctively, and the furthest away from being stored or replicated digitally,"